Preliminary Design Tool for Flying Wing Unmanned Arial Systems

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Abstract

Flying Wing Tool is a software environment for the preliminary design of tailless aircraft built around the open-source AeroSandbox ecosystem and extended with custom geometry, structural, mission, and CAD export capabilities. The motivation is simple: flying-wingdesign is highly coupled, but the preliminary workflow is often fragmented. Geometry is defined in one place, aerodynamics in another, mission assumptions in a third, and export in a fourth. The result is slow iteration, weak traceability, and repeated opportunities for the aircraft definition to drift. The implemented system addresses this problem by organizing the design process around a shared project model. From that common representation, the software supports parametric geometry definition, spanwise section generation, aerodynamic analysis, wingbox-oriented structural assessment, propulsion and mission evaluation, and export to downstream CAD and manufacturing formats. The contribution of this work is a coherent preliminary-design workflow that keeps these capabilities connected.

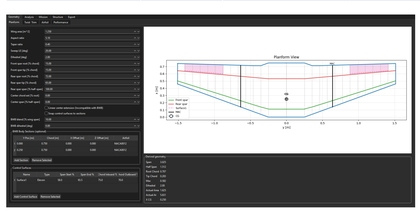
Research Gap and Contribution

- Flying-wing design is tightly coupled: planform and twist changes affect trim, loading, structure, and manufacturability together.
- Fragmented tool chains slow iteration because geometry and assumptions are repeatedly re-entered.
- The main contribution is software integration: one project state supports geometry, aerodynamics, mission, structure, and export.

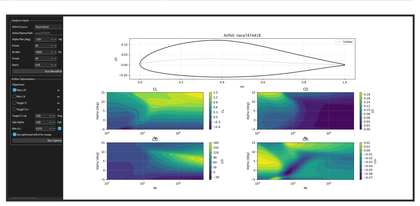
Implemented Workflow

- GUI tabs: Geometry, Analysis, Mission, Structure, and Export.
- Shared state: one project object stores wing definition, trim assumptions, mission settings, and outputs.
- Service layer: geometry or aerodynamics, mission or propulsion, wingbox-oriented structure, and export.
- Reproducibility path: run_full_analysis.py reloads saved projects for repeatable reruns.

Module	State	Role in the current thesis scope
Geometry	Active	Parametric planform, airfoil, control-surface, and section generation
Analysis	Active	Aero polars, twist synthesis, and performance metrics
Mission	Active	Stored mission simulation and phase-level summaries
Structure	Active	Wingbox sizing, margins, deflection, and feasibility checks
Export	Active	CAD and manufacturing-oriented geometry handoff
Batch rerun	Active	File-based reproducibility from saved projects



Geometry tab for planform, airfoil, and control-surface definition.



Analysis tab for aerodynamic sweeps and post-processing.

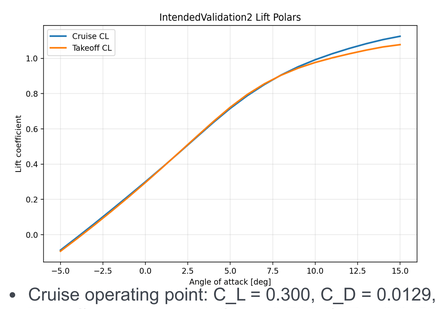
Why the Integrated Loop Matters

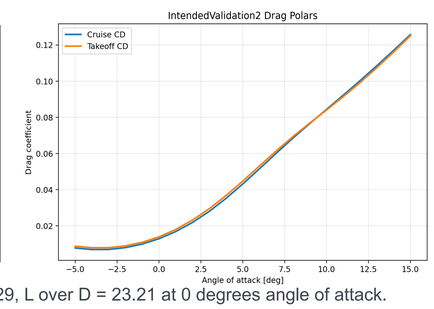
- The same spanwise sections drive aerodynamic loading, structural sizing, and exported geometry.
- Twist optimization is written back into the same project state used for later structural and mission analysis.
- One saved case can expose aerodynamic promise together with a governing structural or drag limitation.

Representative Case: IntendedValidation2

Planform area	1.625 m^2	Cruise speed	19.95 m/s
Span	3.025 m	Takeoff speed	13.34 m/s
Aspect ratio	5.63	Cruise angle of attack	0.00 deg
Gross weight	12.0 kg	Takeoff angle of attack	5.09 deg
Lift target	Bell-shaped	Control surface	Single elevon set
Static margin	8%	Centerbody type	Blended body sections

Aerodynamic and Twist Results





- Cruise operating point: C_L = 0.300, C_D = 0.0129, L over D = 23.21 at 0 degrees angle of attack.
- Takeoff operating point: C_L = 0.730, C_D = 0.0456, L over D = 16.02 at 5.09 degrees angle of attack.

Integrated Structure and Mission Outputs

Structure

```
Structural mass       2.673 kg
Tip deflection        2.3 mm
Tip twist             0.197 deg
Lowest Factor of Safety: Rib Crushing
at 1.6
Threshold Factor of Safety 1.5
Feasibility flag      True
```

Mission

```
Mission completion    9 of 9 phases
Distance             1283 m
Energy used          6.64 Wh
Final SOC            78.1%
Max electrical power 2.30 kW
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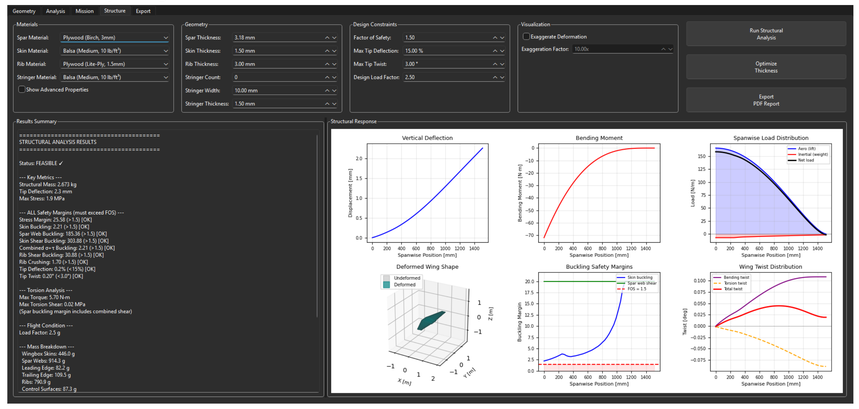
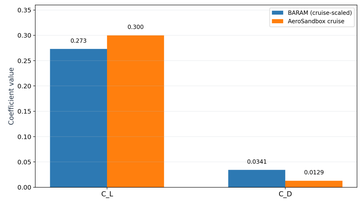


Structural-analysis interface used to inspect stress, margins, and feasibility for the same

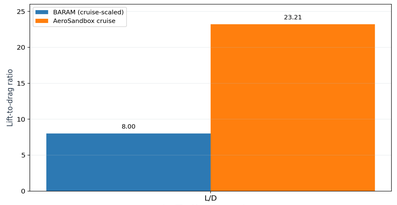
Case Study Interpretation

The case study shows why the shared-state workflow matters. The same saved aircraft produces a coherent cruise solution, a usable twist distribution, a lightweight structure, and a mission-level energy summary. The tool therefore surfaces coupled design consequences rather than isolated outputs.

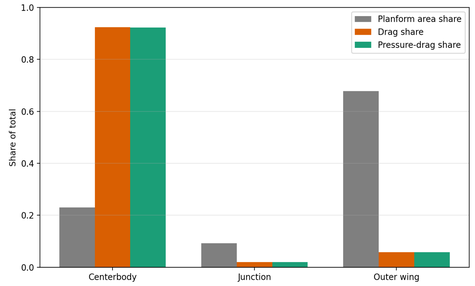
Validation: OpenFOAM CFD Cross-Check



coefficient comparison for C_L and C_D.



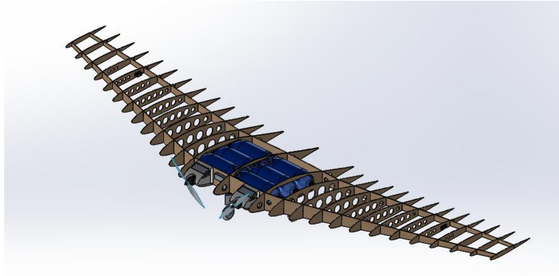
L/D Comparison



Projected planform share versus split-patch drag share.

- Independent CFD path: incompressible RANS, standard k-epsilon, symmetry half-model, convergence after 604 SIMPLE iterations.
- After rescaling to the thesis reference quantities, lift is directionally close: C_L = 0.273 in BARAM versus 0.300 in the repository cruise analysis.
- Drag differs materially: C_D = 0.0341 in BARAM versus 0.0129 in the repository cruise analysis, reducing L over D from 23.21 to 8.00.
- The CFD drag is pressure dominated: 77.2 percent pressure drag and 22.8 percent viscous drag.
- The centerbody plus junction occupy only 32.2 percent of projected planform area but account for 94.3 percent of the drag share, indicating a centerbody-dominated pressure-drag mechanism.

CAD Model



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